

100% FREE & OPEN ACCESS

Complete AI & Data Science Mastery

Lead Instructor: **Golam Maula Lincoln** | github.com/gmlincoln

An End-to-End Industry Roadmap: From Python Fundamentals to Generative AI & MLOps

28+

CORE MODULES

100%

FREE FOREVER

10+

PROJECTS & LABS

Production

MLOPS & CLOUD

COMPLETE COURSE CURRICULUM BREAKDOWN

1. Foundations & Programming

- **Python:** Syntax, OOP, Data Structures, Lambdas
- **Data Pre-processing:** Cleaning, Scaling, Encodings
- **Data Visualization:** Matplotlib, Seaborn, EDA
- **SQL for Data Science:** Queries, Joins, Aggregations

2. Mathematics & Core ML Concepts

- **Math for AI:** Statistics, Linear Algebra, Probability
- **Evaluation Metrics:** Precision, Recall, F1-Score, ROC
- **Bias/Variance Tradeoff:** Overfitting, Regularization
- **Machine Learning Fundamentals:** Paradigms & Workflow

3. Supervised & Unsupervised Learning

- **Supervised:** Regression, Classification algorithms
- **Logistic Regression & SVM:** Decision boundaries, Kernels
- **Decision Trees & Ensembles:** Random Forest, Boosting
- **Unsupervised:** Clustering, Association Rules, PCA
- **K-Means Clustering:** Centroids, Elbow method

4. Reinforcement Learning

- **Reinforcement Learning:** Agents, Environment, Rewards
- **Q-Learning & MDP:** Policy optimization basics

5. Deep Learning & Architectures

- **Perceptron & Neural Nets:** Activations, Backprop
- **Architectures:** FNN, CNN (Vision), RNN/LSTM (Sequential)
- **TensorFlow & Keras:** Model training, Pipelines
- **Generative Adversarial Networks (GAN):** Generators & Discriminators

6. Generative AI, LLMs & Advanced NLP

- **NLP Foundations:** Tokenization, Embeddings, TF-IDF
- **Transformers:** Self-Attention, Encoders, Decoders
- **GenAI & LLMs:** Prompt Engineering, Fine-tuning
- **OpenAI APIs:** Function Calling, Embeddings, Systems
- **RAG (Retrieval-Augmented Gen):** Vector DBs, LangChain

7. MLOps & Production Engineering

- **Flask:** Building Web APIs for Model Serving
- **Git & GitHub:** Version Control, PRs, Workflow
- **Docker:** Containerizing AI microservices
- **Kubernetes:** Scaling & Orchestrating AI containers

8. Projects & Capstone Experience

- **Multiple Minor Projects:** Sentiment, Churn, Vision
- **Major Enterprise Projects:** RAG Agent, Full-Stack GenAI Web App & K8s Pipeline

Course Author: **Golam Maula Lincoln** | GitHub: github.com/gmlincoln | Open Source MIT License